

Decomposition of Policies for Cluttered Scene Grasping

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Abstract—In cluttered-scene robot manipulation tasks, a manipulator needs to avoid obstacles and pick up a target object. While analytical motion and grasp planning methods often build on top of 3D geometry and accurate state estimation, model-free RL methods mostly focus on top down grasping. How to map perception to more complicated 6 Dof control remains a challenge. In this work, we study an end-to-end system that decomposes the tasks into two components: an imitation agent and a model-free reinforcement agent. Moreover, we present some preliminary results on building these controllers and ideas to combine them into a coherent framework. We compare different representations for an imitator to leverage the structure of the tasks and abundant simulator demonstration data to generate effective obstacle avoidance. We also experiment with different algorithms for the RL agent to grasp a target object from hand camera view.

Index Terms—Reinforcement Learning, Grasping, Imitation Learning, Optimal Control, Vision, Neural Networks

I. INTRODUCTION

In order to manipulate objects in a cluttered scene, a robot often relies on planning algorithms to generate motion to approach a target object in the scene. This often requires perfect state estimation and knowledge about the workspace geometry. Moreover, it requires a low-level controller to execute the plan and react to the environment.

Many seminal trajectory-optimization papers have shown strong candidates for the solutions by hand designing such a cost [35, 37, 42]. Recent works on reinforcement learning have proposed to learn continuous manipulation tasks that have complex contacts and dynamics through trials and errors in simulators [27, 43]. Works that apply in real-world robots often build off large-scale dataset collections [13, 17] and efficient model-based methods [16, 42]. In general, these algorithms often require a large amount of environment interactions and are hard to train, especially for long-horizon tasks with sparse rewards. Specifically for vision-based grasping, many papers [13, 26, 33] focus on top-down grasping with cluttered targets, due to its tolerance for large exploration and connection to computer vision.

In this work, to overcome the limitations of existing methods, we propose to combine supervised learning for obstacle avoidance and reinforcement learning for robust grasping. This is motivated by the industry practice to accomplish manipulation in cluttered scene. The system is often divided into two components with two cameras: An overhead camera monitors



Fig. 1. Our Framework composes of two components: An imitation learning agent for collision avoidance and a reinforcement learning agent for precision grasping. A consistent policy would be coherently “generated” from these two components.

the obstacles in the scene and predicts an approximate end effector pose to reach the target and a wrist camera that takes charge of precision grasp when target object is inside the field of view. With sufficient large amount of data, classical machine learning theory gives a strong foundation for the generalization of our collision avoidance in configuration space. Meanwhile, model free RL methods in task space with off policy data are more efficient to learn and likely transferable into the real world. In this paper we address the progress of the projects including some formulations and experiments, and open up future directions to continue approaching the problem. To summarize, we present (1) Learning obstacle avoidance from demonstration. We compare image representation and our point-SDF representation for this tasks. (2) Learning precision grasping with model-free RL. We experiment with different algorithms and analyze their performance. (3). We propose ideas for integrating an end-to-end grasping system and some next steps based on the current progress.

II. RELATED WORKS

A. Model-based Approach

Model-based control methods assume some knowledge about the dynamics (model), and often apply stochastic optimal control algorithms such as iterative Linear Gaussian Policy (iLQG) [37], Path integral for Policy Improvement (PI2) [35] and Model Predictive Path integral (MPPI) [42] or Model Predictive Control (MPC) approaches such as [7, 30]. Specifically for modelling the contacts, [20, 25, 38] have shown impressive trajectory optimization in simulators and real world. On the other hand, one can fit dynamics with collected data, and apply function approximation such as a neural net to model the dynamics. [6] uses a deep net to generate visual foresight for manipulation tasks. Recently [41] benchmarks model-based controllers in simulation domains.

While it's not trivial to acquire a model of the real world dynamics, these methods are superior in data efficiency and interpretability.

B. Model Free Reinforcement Learning

Numerous works have adopted deep reinforcement learning approaches for manipulation skills. [13, 24] respectively builds on Deep Deterministic Policy Gradient (DDPG) and Deep Q-learning (DQN) to learn general and robust control policies. [27, 43] uses demonstration data to circumvent the need for reward shaping often used to guide exploration for policy gradient methods. [28] benchmark the performance of different continuous control algorithms including Trust-Region Policy Optimization (TRPO) [32] and Natural Policy Gradient (NPG) [12] with linear approximation and RBF kernel and a focus on diverse trajectories. Model-free methods remove the bias in modeling the world and sometimes have better asymptotic performance with a universal algorithm [11]. However, even with state information, these approaches still require millions of interactions to learn a single task and exhibit large variance to hyper-parameter and reward design.

C. End to End Vision-based Robot Manipulation

Many learning-based methods perform robot manipulation directly from high dimensional input such as RGB images. [16] propose to use Guided Policy Search (GPS) for end-to-end visuomotor control. [26] benchmark various off-policy methods for top down grasping in simulators. [17, 23] makes use of large-scale self-supervision data to learn vision-based grasping. [21] deploy closed loop grasping pipelines based on grasp detection algorithms. While a static RGB sensor is often hard to reason about the 3D complexity of the manipulation tasks and found poor generalization, depth point clouds and voxel often require nontrivial registration and calibrations. Some recent works [4, 8] use active learning to determine successive sensor placements to improve state estimations. [2] use multiple sensors to integrate more information in states to synthesize controls. More recently, [33] use fusion as model to generate action view, and efficiently train a Q network for closed-loop grasping. Overall, end-to-end control systems have achieved some impressive results on manipulation tasks. These grasping papers mostly consider operational space control for top-down grasping in simulation domains, whereas in real worlds, for instance a kitchen robot, It's desirable to have 6DOF grasping and collision avoidance in cluttered scenes.

III. METHODOLOGY

Consider in a human-centric environment where a manipulator is asked to grasp a target object from a table without colliding with drawers, tables, and other objects. This task is much more challenging than top down object picking and often require a planner to generate grasping trajectories and a robust controller to do path following. Due to the complexity of this task, purely RL methods from observation may take infinite time to converge to a good policy in simulations, not to mention generalization to various scenes and robustness to

dynamics noise. Therefore, we propose to divide this complex problem into two subtasks: collision-free target reaching and dexterous object grasping. The idea is to leverage the abundant demonstration data to learn obstacle avoidance and use model-free RL methods to learn precision grasping.

Formally, We model the control problem as a Markov decision process (MDP), which is defined using the tuple: $\mathcal{M} = \{\mathcal{S}, \mathcal{R}, \mathcal{A}, \mathcal{T}, \rho_0, \gamma\}$. $\mathcal{S}$ and $\mathcal{A}$ represent the state and action space. $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function. For most top-down grasping tasks, the reward is an indicator function given at the end of the episode denoting if the target object is lifted and there's no collision. $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the transition dynamics. In model-free RL, we are only allowed sample access to this function. ρ_0 is the probability distribution over initial states and $\gamma = [0, 1)$ is a discount factor. Our goal is to solve for a policy of the form $\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_+$ which optimizes the expected returns:

$$\eta(\pi) = \mathbb{E}_{\pi, \mathcal{M}} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (1)$$

We also define value, Q, and advantage functions associated with a policy as follows:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{\pi, \mathcal{M}} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right] \\ Q^\pi(s, a) &= \mathbb{E}_{\mathcal{M}} [R(s, a)] + \gamma \mathbb{E}_{s' \sim \mathcal{T}(s, a)} [V^\pi(s')] \\ A^\pi(s, a) &= Q^\pi(s, a) - V^\pi(s) \end{aligned}$$

The main challenges different from top-down grasping is that the object can fall down with an inappropriate touch, and the camera might easily lose the views during exploration. Moreover, our experiments show that including shaped reward designs such as collision, target falling down, reaching the target would make policy improvement less stable. Therefore, we only include an episode pretermination if the target object falls down.

A. Imitation Learning (IL) for Obstacle Avoidance

Model-free RL methods can take many samples to learn a policy due to the exploration of the state-action spaces, which can possible be task irrelevant. To combat this, our first thought is to use behavior cloning (BC) from existing motion and grasp planner [40] to provide an informed policy initialization that efficiently guides exploration. Denote our collected dataset from expert rollout as $p_D = \{(s_t^i, a_t^i, s_{t+1}^i, r_t^i)\}$. As a supervised learning procedure, BC has the follow maximum log likelihood objective

$$\max_{\theta} \sum_{s, a \in p_D} \ln(\pi_{\theta}(a|s))$$

BC alone suffers from the distributional shift between the demonstrated states and the policy's own states [31]. In our setting, the multimodality of grasping and the size of the state space also make BC hard to generalize. In RL training [27, 43], an imitation loss is often combined with task reward with a weighting schedule.

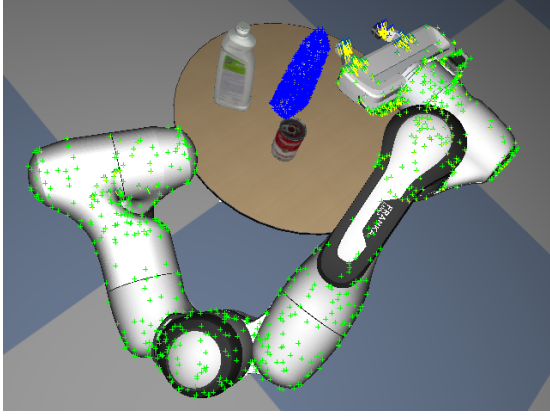


Fig. 2. Point Cloud Representation

Specifically, we denote the manipulator's joints $q \in \mathcal{Q} \subset \mathbb{R}^d$ where d is the degree of freedom and end effector pose as $x \in \mathbb{SE}(3)$. While often images are used as high dimensional representation, it can have problems in sim to real transfer. So we adopt a set of sampled points on robot as our observation and at each time step, we compute the robot points in work space by forward kinematics, denoted as p_{robot} . Moreover, we specify the target object as segmented observed point clouds p_{target} . We denote this imitator agent as π_{obs} .

Since the scene point clouds can be complicated and contain irrelevant information, we make an assumption that the Signed Distance Field (SDF) of the scene can be computed, which may come from fusion or pose estimation of the known objects. To minimally encode information about the environment objects other than target, we compute the signed distance field gradient of the robot points p_{robot} as α_{robot} , which provides information for the policy to move away from the obstacles. The state for collision avoidance is represented as $s_t^{\text{col}} = (p_{\text{robot}}, \alpha_{\text{robot}}, p_{\text{target}})$, as shown in Figure 2. We represent the policy as a neural network and the control is defined as the Δq which then applies to the current joints $q^+ = q + \Delta q$. With our current implementation, we observe a more stable performance from supervised learning without finetuning with model-free methods such as PPO.

B. Reinforcement Learning (RL) for Standoff Grasping

In practice, a planner often has trouble optimizing colliding with target object while performing precision grasps. A workaround is that the controller switches to a ‘‘grasping’’ mode from a standoff pose (defined as a short distance from the target in gripper facing direction). Motivated by this trick, we consider a grasping task that requires robot to move in workspace from gripper's local coordinate (ego-motion) by taking in hand camera images, which is similar to top down grasping. Our state representation for precision grasp denoted as s_t^{grasp} is hand camera image, as shown Figure 3 scaled to $64 \times 64 \times 3$. The control is defined as the end effector position in task space $x^+ = x + \Delta x$. We denote this reinforcement learning agent as π_{grasp} .

We would introduce common deep RL methods that can work with off-policy data: soft actor critic (SAC [10]), Double Deep Q Network (DDQN [39]) and Monte Carlo Return Estimate (MCRE [34]). DQN aims to estimate the Q-function by minimizing the Bellman error, given by

$$\mathcal{E} = \frac{1}{2} \mathbb{E}_{s,a} [(Q_{\theta}(s,a) - (r(s,a) + \gamma \max_{a'} Q_{\theta}(s',a'))^2)] \quad (2)$$

To reduce the effects of overestimation in the next states introduced by bootstrapping, DDQN introduces a target network to decouple the correlation, resulting in the following error estimate:

$$\mathcal{E} = \frac{1}{2} \mathbb{E}_{s,a} [(Q_{\theta}(s,a) - (r(s,a) + \gamma Q_{\theta'}(s', \arg \max_{a'} Q_{\theta}(s',a'))^2)] \quad (3)$$

Monte Carlo Return Estimate tries to minimize the loss for Monte Carlo policy evaluation:

$$\mathcal{E} = \frac{1}{2} \mathbb{E}_{\tau \sim \pi} [\sum_t (Q_{\theta}(s_t, a_t) - (\sum_{t'=t}^{\infty} \gamma^{t'-t} r(s_{t'}, a_{t'}))^2)] \quad (4)$$

This resembles supervised learning and requires no bootstrapping. However, it's only unbiased when we use on-policy data. These value estimation method with function approximation are used by many top-down grasping frameworks [13, 17, 33]

The introduction of an actor makes policy optimization more straightforward and is often found to work in continuous control settings. Soft Actor Critic (SAC) adopts max entropy reinforcement learning, where the augmented policy objective becomes

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} [\sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) + \alpha H(\pi(\cdot|s_t)))] \quad (5)$$

and $H\pi(\cdot|s_t)$ denotes the entropy of the policy at that state and α is a trade-off coefficient that can be solved or optimized. The Bellman equation for Q^{π} is then defined as

$$Q^{\pi}(s,a) = \mathbb{E}_{\mathcal{M}} [R(s,a)] + \gamma \mathbb{E}_{s' \sim \mathcal{T}(s,a)} [V^{\pi}(s') + \alpha H(\cdot|s')] \quad (6)$$

The definition for Q^{π} and V^{π} should also be changed accordingly. The entropy regularization helps the policy to explore the state space and found more robust to noises and tasks with multimodality.

C. End-to-End Visuomotor Framework

To fully achieve the task of cluttered scene grasping, we want to integrate two policies π_{obs} and π_{grasp} seamlessly into a policy π_{int} . An augmented representation of the states would be $s_t^{\text{int}} = (s_t^{\text{col}}, s_t^{\text{grasp}})$. In this section, we discuss a few plausible future directions. The advantage of end-to-end system lies on its performance and consistency, which can be critical for multimodal and long-horizon tasks. Composition of policies provide advantages such as reusability and quick initialization for an end-to-end system. For instance, [36] considers combining independent rewards via soft maximization.

Soft Q learning [9] has been shown to compose multiple policies trained for different tasks. Maximum entropy policies



Fig. 3. Hand Camera View

provide an informed exploration strategy which more likely sample higher expected reward actions. While simply adding Q-functions does not generally give the Q function for the combined task, the regret from using the policy by adding the constituent Q functions is upper bounded by the difference between the composed policies [9]. Intuitively, in soft Q learning, each policy has a wider distribution, and therefore the policies can have more overlaps and reduce the regret from composition.

On the other hand, GPS [15, 19] provides a framework to simultaneously optimizing policies while constraining the controllers to be not far away from each other. This constrained formulation can be used to synthesize a single policy that takes in an augmented state space and outputs joint positions as actions. Concretely, algorithm 1 represents a generic guided policy search algorithm [19] for two policies π_1, π_2 . The third step in algorithm 1 is critical because supervised learning might not guarantee that $\pi_2(u_t|x_t)$ will achieve long term performance as $\pi_1(u_t|x_t)$. Similar to [15], trust region updates can be used to constrain policy improvements and policy convergence. Multiple questions immediately arise from inspecting on this idea. (1) The RL agent can only provide supervision for short-horizon grasping. (2) Our expert planner can also be used to provide demonstration data in a short horizon. (3). The large sample complexity for both $\pi_{\text{obs}}, \pi_{\text{grasp}}$ can affect the training stability.

Algorithm 1: Generic Guided Policy Search

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for  $k = 0, \dots, K$  do
  C-step: improve each  $\pi_1(u_t|x_t)$  based on surrogate
    cost  $\hat{c}_i(x_t|u_t)$ , return samples  $\mathcal{D}_i$ 
  S-step: train  $\pi_2(u_t|x_t)$  with supervised learning on
    the dataset  $\cup_i \mathcal{D}_i$ 
  Modify  $\hat{c}_i(x_t|u_t)$  to enforce agreement between
     $\pi_2(u_t|x_t)$  and each  $\pi_1(u_t|x_t)$ 
end

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Safe RL methods such as [1, 22] more explicitly model the collision avoidance as constraints in MDP. However, these

methods are likely to suffer the same sample complexity as normal model-free RL methods. More generally, operational space control [14] can be used to combine multiple policies in task space. [29] introduces Riemannian Motion Policies (RMP) to combine and transform policies in a geometrically consistent way. Mapping our π_1, π_2 to task space as desired goal destination and define the associated metric, we can efficiently use RMP to integrate policies while keeping them intact. Other approaches such as Hierarchical RL [18] models the problems as to combine high-level planner and low-level controllers.

IV. IMPLEMENTATIONS

We experiment with both image-based and point-based representation and different architectures. Our “Single Domain” dataset are collected from planner trajectories [40]. The training dataset contains nearly 250000 state action pairs. The domain randomization dataset augments to 1 million state action pairs. The scenes are generated by dropping sampled YCB objects [3] with random upright poses on a table in the Pybullet Simulator [5]. Each scene samples 3 to 7 objects. The objects used are master chef can, cracker box, sugar box, tomato soup can, mustard bottle, potted meat can, pitcher base, bleach cleanser, bowl, and mug. The target is fixed to be a standing bleach cleanser in “Single Domain” dataset.

A. Obstacle Avoidance

For image-based input, we rescale the image to $224 \times 224 \times 5$ with channels RGB, depth, and mask. In addition to the rich color information, the depth helps with collision avoidance and the mask is used to specify target. Our overhead camera is placed in a fixed position and has overall view of the arm and the obstacles. We use pretrained Resnet-18 as backbone and instead of predicting joint position increments, we directly regress on the target joint position of the next state. For point-based input, since the correlation among points are critical in our representation, our network composed of convolutional layers embedded into 256 latent channels followed by fully connected layers. Our network architecture is composed of convolutional layers embedded 6-channel input into 256 latent channels to extract the features of the collision points. Our target points are also processed through a network with same architecture with 3 input channels. Two features are concatenated before fully connected layers to compute control and supervised with MSE loss as standard behavior clone. In addition, we have added an auxiliary relative position loss between end effector and target object to stabilize training.

B. Precision Grasping

For these off-policy methods, we starts with a replay buffer of experience collected from a heuristics exploration policy. During training, we adopt an epsilon greedy exploration strategy to bias gripper moving forward. For Q network, we apply double-clipped Q learning with two target networks and replace the standard MSE loss with cross entropy as in [13]. We also experiment with prioritized experience replay and clip

Single Domain / Randomized Domain		
Metric	Image	Point-SDF
Grasping Success	0.73/0.39	0.36/0.32
Collision Avoidance	0.94/0.92	0.97/0.95
Reaching Performance	0.96/0.92	0.78/0.73

TABLE I
IMITATION LEARNING PERFORMANCE

the gradient in the network updates. For SAC, we parametrize the policy as Gaussian policy with fixed variance. We also experimented with For all methods, we collected 30000 transitions in the replay buffer before on-policy experience. We run 5 update steps with a batch size of 256. In evaluation time, we adopt the derivative-free cross-entropy method (CEM) to select actions for DDQN and MCRE. Our network contains convolutional layers embedded input RGB images into 128 channels and followed by connected layers to predict the position control.

V. EXPERIMENT

In this section, we discuss some preliminary experiments on the performance of collision avoidance controllers and grasping controllers.

A. Imitation Learning Ablation Study

While an expert planner has full object geometry and poses for motion planning, our learned policy will map from raw perception to controls. Image representation has been adopted in many RL papers and demonstrated various kinds of success due to the development in computer vision field. In our supervised-learning experiments, the image-based methods have more promising behavior cloning, which comes from the representation power of network. In contrast, our point and SDF representation structurally encode the target and obstacle information and generalize better in various domains. Since the goal of the imitator is to avoid collision and reach object, we do 100 rollouts of the trained policy on new scenes and evaluate in three metrics: grasping success denotes the percentage of episodes that target is lifted at the end of episode, the trajectory is collision-free, approximate reaching denotes the fingertip center is within 0.2 meter to the target object center.

Overall, we observe that image-based representation have a better performance for grasping success, and point-based representation performs comparably in obstacle avoidance. We suspect the image provides richer information to reason about grasping. However, we expect simulation images, including color, depth, and mask to be different from real world in many aspects. Moreover, we are willing to assign accurate grasping to reinforcement agents. Note that it's critical to provide pre-trained backbone for image-based imitation learning, without that both networks perform relatively the same. Overall, both representation has a good performance of reaching if trained with sufficiently diverse state-action pairs.

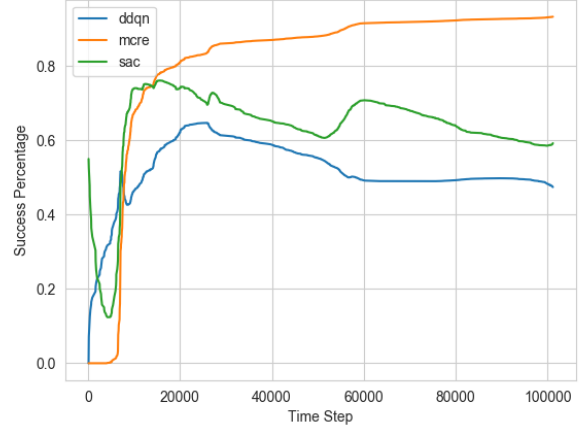


Fig. 4. Training Reward Comparison

B. Reinforcement Learning Ablation Study

We evaluate the performance of off-policy methods that can make use of collected data. Overall, DDQN and MCRE perform relatively better than SAC which has an additional actor and often perform very conservatively. Empirically we found that training MCRE is more stable than the others. Due to the complexity of contacts for grasping, we observe that these algorithms with pure off-policy data usage are not capable of reaching similar performance as on policy. Moreover, over multiple runs, DDQN turns out to converge quickly and then become unstable. We suspect this can come from the overestimate of the values and lose target views during training. We tune different hyper-parameters and training schedule for SAC and other on-policy methods with an actor for continuous control, and the policy is still not able to reach the same performance based on observation. Therefore, we fix the feature network for SAC and gradually increase the training horizon.

VI. DISCUSSION AND FUTURE WORKS

While the idea of decomposition and multiple policy come from the challenges to train a single policy for cluttered scene grasping, we are exploring ideas to have a more coherent representation and learning objectives. For imitation learning, we need more interpretation and experiments to verify the point-SDF structure. For image-based representation, while the network starts to struggle with domain randomization, we are also exploiting more structures of the object, environment interactions for better learning.

The immediate next step for RL framework is to include orientation in our action space, which can be represented as Euler angles. Moreover, similar to [33], we find that supervised learning approach improves and stabilizes the performance, so we will experiment with expert demonstrations and diverse environments. We will also experiment with pixel-wise value prediction and action views for Q learning in addition to a supervised learning initialization. The dense Q prediction

backprojects the success observation to the trajectory, which acts as a attention mechanism. The dense predictions provides more supervision for stability. Moreover, action-view prediction with two towers structures are applicable if fusion are adopted and should be easier to learn, compared with separating state and action representation.

Overall, we have been following our plan to mostly exploit the limits of standard supervised learning for those tasks. The future works would mainly focus on the generalization performance of π_{obs} and π_{grasp} . More importantly, we can develop ideas to improve and integrate both policies for a coherent framework. Note that while we experiment with simulators and make use of on-policy data, we expect to have efficient data usage for real-world finetuning purposes.

VII. CONCLUSION

In conclusion, we have presented a control project specific for cluttered scene grasping tasks and some preliminary experiments, with a focus on generalization in real world and robustness. While the task of visuoservo is challenging, we decompose the system into collision avoidance and precision grasping. We make use of simulation demonstrations for pretraining and supervised learning. We believe that there are some interesting future directions to continue the research.

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